

Senior Project: COINS Network Overhaul

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Senior Project

Summary

The idea for this project came from necessity. I was working as a lab assistant doing all of the windows updates on the computers. This would take a long time just to do some windows updates and was quite annoying. We were doing windows updates from Microsoft.com which is the way windows is designed for the average user at their house. However we are using a domain with a server so there had to be an easier way. My goal was to find that way and I learned a lot about how to make a server accomplish all of the things I needed to do on each machine. It ended up being a COINS Infrastructure Enhancement Project.

Infrastructure Enhancements

1. DNS Server

The first problem we had was the DNS server on our server. The DNS was set to loop back on itself which is not good. We had to change this because the internet was taking a long time to return a web page. Now we perform much faster web page look ups. Then we had to set all of the client machines to the correct DNS server addresses. They needed to have 192.168.0.2 as the primary DNS server and 192.168.0.1 as their secondary DNS server. This makes the clients look to the server for their global policies and logon scripts.

2. WSUS Server

Installing the WSUS Server Role was the next step. To install the WSUS Server you open your server Manager and go to the Roles menu. Once you are in that menu go into add roles and select Windows System Update Services (WSUS) and take all of the defaults. If it requires you to add more features to make this Role Work then install them as well. Once the role installs you must restart the Server for the Role to take full effect. Now WSUS is installed and running. Make sure all of the role features are turned on by going into your server manager and now you will see a WSUS tab. Select it and on that

come up make sure all features are running. It will prompt you to see if you want it to go get updates from the default Microsoft site or another WSUS server. For our environment we want it to go to the default Microsoft site. Now start the download and wait a few hours for it to pull in all of the update packages.

3. Default Global Policy Set-up

Now I had a WSUS up and running. It was time for the Global Policy to be set up. Open the group policy editor and drill down through the menu on the right to the default domain policy. The default domain policy is for all of the windows computers on the network. Typically you do not edit the default policy but for WSUS you can make an exception. All of the machines need to check into the WSUS server to get their updates. You right click the policy and then make sure “enforced” is checked so this is the policy they are using and then you click edit. In the next screen open windows settings, administrative templates, and then scroll down to Windows System Update Services. The Fourth Setting is set to disabled. It is specify a Microsoft intranet site update location. Set it to enabled and put in the name of the Server. For COINS department the name of the Server is <http://server.COINS.csuniv.edu>. Clicks apply to save these changes to the policy. The other ones that I changed were the wake on LAN, the times that the computer would check in for updates, and the enable automatic restarts. The clients typically check-in for domain policies every 12 hours. You can wait the 12 hours for the clients to check in or go out to each client and run the command from command prompt “gpupdate /force”. This forces the client to go and apply polices. Now if you open the server manager and go into your WSUS Role and drill down to the computers and click all computers you will see the clients that are coming in looking for their updates. Under all computers you can create a group by right clicking and then naming the group. Go through the menus and select which clients you want to go into each group. I made 7 groups each containing 6-8 computers.

4. Create Computer Groups in Active Directory

The next step was to group the computers in Active Directory. Open the Server Manager and go to the Active Directory Role drill down through the domain and down to the computers folder. Once you are in the folder, right click and click add and say group. Using the menus or drag and drop put the entire computer you want in each group. I made 4 groups 1 for the macs and 3 for each room and dropped all of the machines into their respective groups.

5. Create Organizational Units (OU's)

To Create an OU go into the Server manager and select go under active directory, users and computers. Once you are there you can right click and say new Organizational Unit. I created 4 OU's, one for each room and one for the MACs. Once you create the OU's you can put groups into OU's. So I put the computer groups in their respective OU's. You cannot put a computer or user into an OU; you can only put the groups into the OU. OU's make management of clients and policies very easy as you will see later.

6. Set up OU policies

To set up OU policies all you have to do is open your group policy editor. Now under your default domain policy you should see the OU folders. Select one and right click and select new policy. Now this policy will only be applied to those machines. They are an enhancement of the default domain policy so you do not have to worry about setting up WSUS policies again because policies work top down. You can allow something in the default domain policy but not allow it in the group policy as you will see later. However, if something is not allowed in the default domain policy however even if it is allowed in the lower policy the highest level has the effect.

7 Network Policy and Access Services (VPN/RADIUS Server)

To install this Role on the Server all you have to do is install it just you did the WSUS server go to roles and add role and select network policy and access services, take all the defaults and wait for it to install. Once it has installed you go into radius clients and add the router or client you would like to provide

radius access to. I added 192.168.0.1, this is our cisco router. Now the server is looking for the router. You feed it the shell privilege level 15 so the router knows the server is allowed admin right. Now you set the network access policy so that they are exchanging the radius packets and now you set the access rights to what machines you would want a vpn user to have access to. For COINS, the only access a user gets is to the server. They cannot remote start a remote session without the administrator username and password however they can access their personal X: drive and public Y: drive. Only the proper credentials from active directory will let them in and so far we have had a 0% security breach.

8. Print Serve

Adding Print and Document services to the Server is just like adding any other role to the server. You install the server on the printer. This is easy most of the drivers automatically download. If you need to download the driver you can do so and install it. Now the Server knows about all the printers on your network and has access to the documents queue and ink levels and error messages. From here with the OU's and policies you can only allow certain clients to install and use certain printers and the clients must use the driver that you provide to them. So machines in 203 can only print to the printer in 203 using the driver that works on all of them. This makes administering where people print and the drivers very easy.

9. Local Update Publisher

The idea of the Local Update Publisher is to give you the maximum flexibility of your WSUS server. It allows you to make and push your own updates. You can download Local Update Publisher from SourceForge. It has everything you need to fully enhance your updating capabilities. You first have to run an update from Microsoft on your WSUS server to let the LUP work with it using its clients and policies. You create a certificate when you open the program and add that to the trusted publishers root in your MMC. Then you go into your default domain policy and put it in under computer settings, trusted pushers. Now when clients check in they will get the key and will allow this update to run. Next

you download the MSI file the program that you want to install. For me it is Mozilla Firefox. I download the MSI file and go to tool, create update. I fill in the distributor and what version it is and then take all the defaults and finish. It crates the update and the package and publishes and I push it to all the clients. Once this is done the clients will download that update on our local network and run it just like any other update and this can be done with Flash as well, and almost any other MSI.

10. Fiber-Optic change over

We switched the Department over to our own fiber-optic connection. I was quite hard and we had to get Tony's help to get the connection to AT&T up and running. Then you hook a fiber optic patch cable into the panel on the wall and hook the other end into the fiber optic converter box. From the converter box we hook the converter with and Ethernet cable to Ethernet 0/0 port on the router. Next using the console port (Ethernet 0/1) on the router and set our computers ip-address to the 192.168.1.XXX subnet with a 255.255.255.0 subnet. Pick any address you want within that subnet. Next open Cisco Configuration Professional and discover and connect to the router. Now click the configure button and the menu on the left will populate. Under router you select the interface gigabitEthernet 0/0 and click manage. Next you click edit and set the ip-address to 97.167.185.3. This is public ip-address and set the dns server to AT&T dns servers and gateways. From config pro use the pig command and ping www.google.com. This will test to make sure you are connected. Now hook gigabitEthernet 0/2 into your switches and now the internet is back up on our fiber connection.

Inventory

I did a full inventory audit of all of the machines we have in our environment. It started by getting the ip-address of each machine and the machine name. Next I got the ip-address and hostnames from the printers and/or switches that had ip-addresses. The hardest to find was the serial numbers on the switches because they are on the bottom and are hard to find. We took the inventory for our A-bit accreditation as well as for depreciation and insurance purposes.

Problems

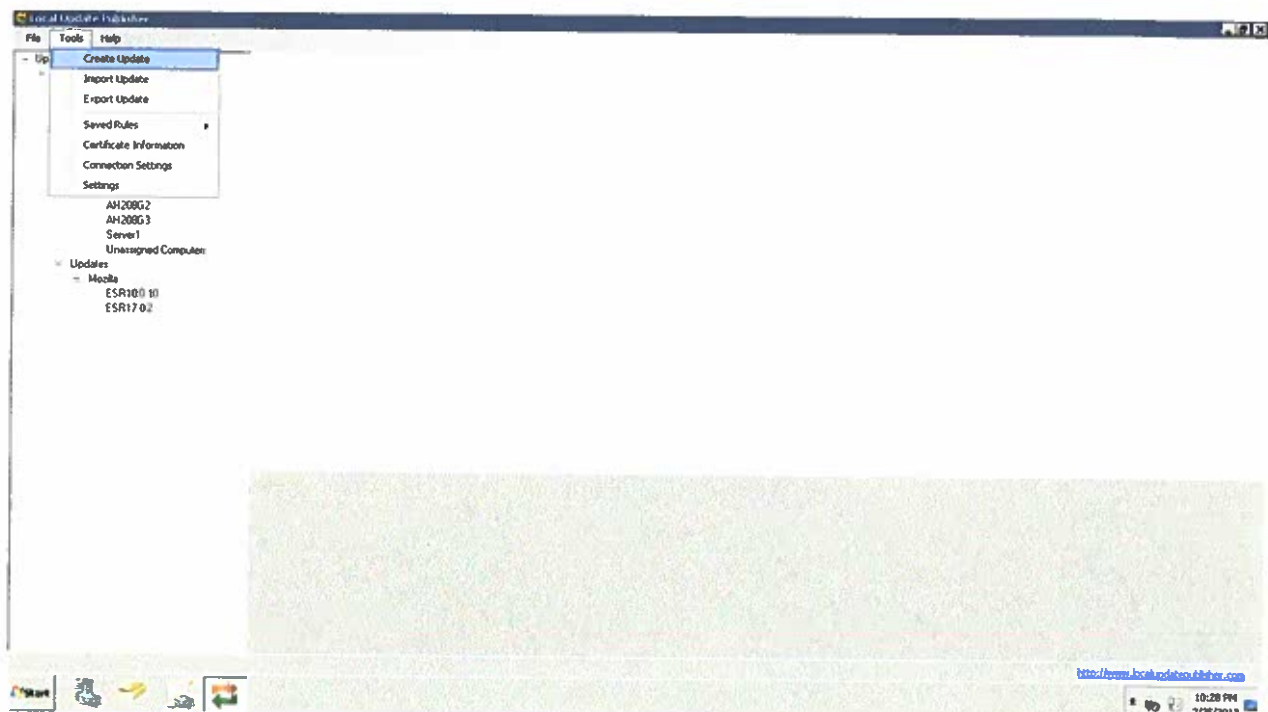
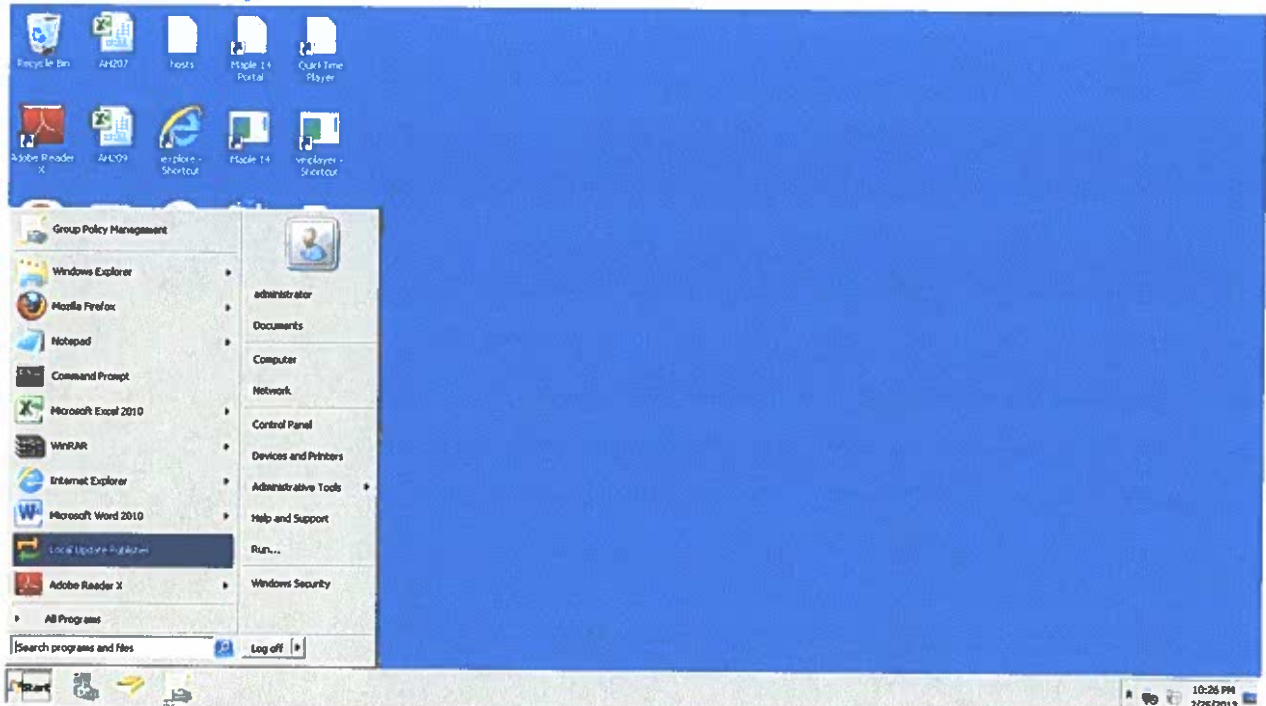
We ran into some major problems along the way. Our third party update server was not being allowed to publish updates to the file that it needed to. We needed to run an update on the server to open the security of .NET to let us write to that folder. Some other problems came when we tried to get the radius server and the radius client talking and the problem was in the network policies not letting the router and the server talk. The server was dropping the packets until we got the client added correctly.

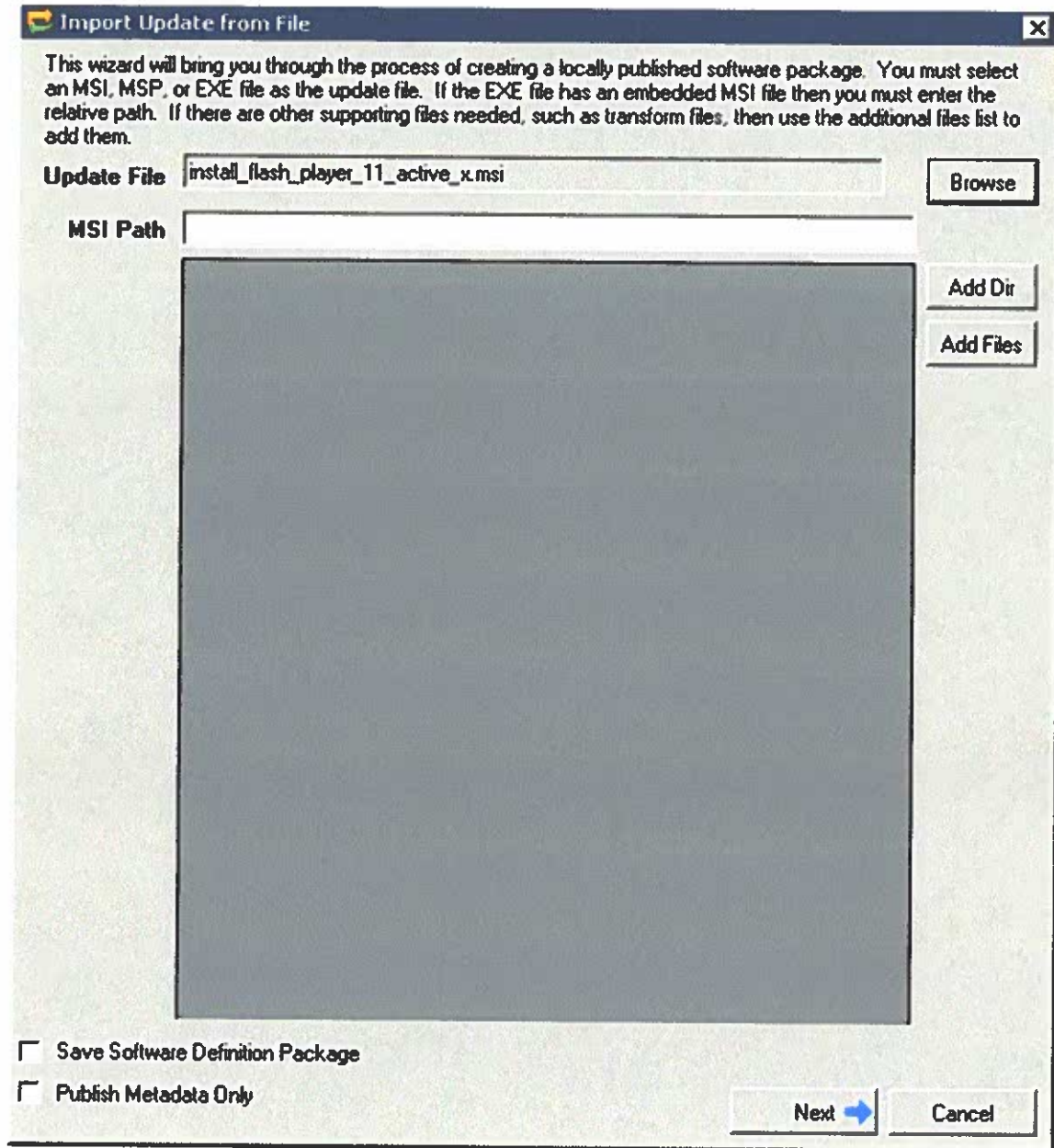
Future Enhancements

This server can handle what we ask of it now but moving forward in the future it should be easy to maintain. The next thing to be done would be to start distributing the environment. If we had a couple more servers one to serve the files and the other to be the WSUS it would take a major load off the server and make a really clean and fast environment. Adding servers to the environment is a Project within itself. It would require multiple migrations depending on what roles the administrator is looking to move. It can take hours for just one migration. And if the migration is not done correctly you have to reformat the new server and start over, not to mention if your switching from Server Enterprise 2008 to Server Enterprise 2012 some things may not work and may not be the same or function the same. Also upgrading some of the switches and possibly putting in a fiber backbone would really help the environments speed and latency.

Using the New Services

Using the Local Update Publisher





Import Update from File [X]

Enter the relevant details about the software package here. Refer to the documentation for questions regarding individual fields.

Package Type	Application
Package Title	Adobe Flash Player 11 ActiveX
Description	Adobe Flash Player 11 ActiveX
Classification	Updates
Bulletin ID	
Severity	Unspecified
Vendor	Adobe
Product	Adobe Flash Player 11
Article ID	
CVE ID	
Original URL	
Support URL	
More Info URL	
Impact	Normal
Uninstall	True
Network Required	False
Reboot Behavior	Can Request Reboot
Command Line	

Prerequisites **Supersedes**

☐ Save Software Definition Package

☐ Publish Metadata Only

[< Previous](#) [Next >](#) [Cancel](#)

Import Update from File

Use this form to create, edit, remove, and group rules that determine if this software package is already installed. If you created this update using an MSI or MSP file you will see the automatically generated rule at the installable item level. By default the rules are in an AND group.

Package Level - Installed Rules

Save Rules Load Rules

Rule

MSI Product Installed {eb08d51e-ba72-46(a-94b2-cba9c(8973d8)}

Add Rule Remove Group Edit Rule Remove Rule

Installation Item Level Edit

<msi:MsApplicationInstalled />

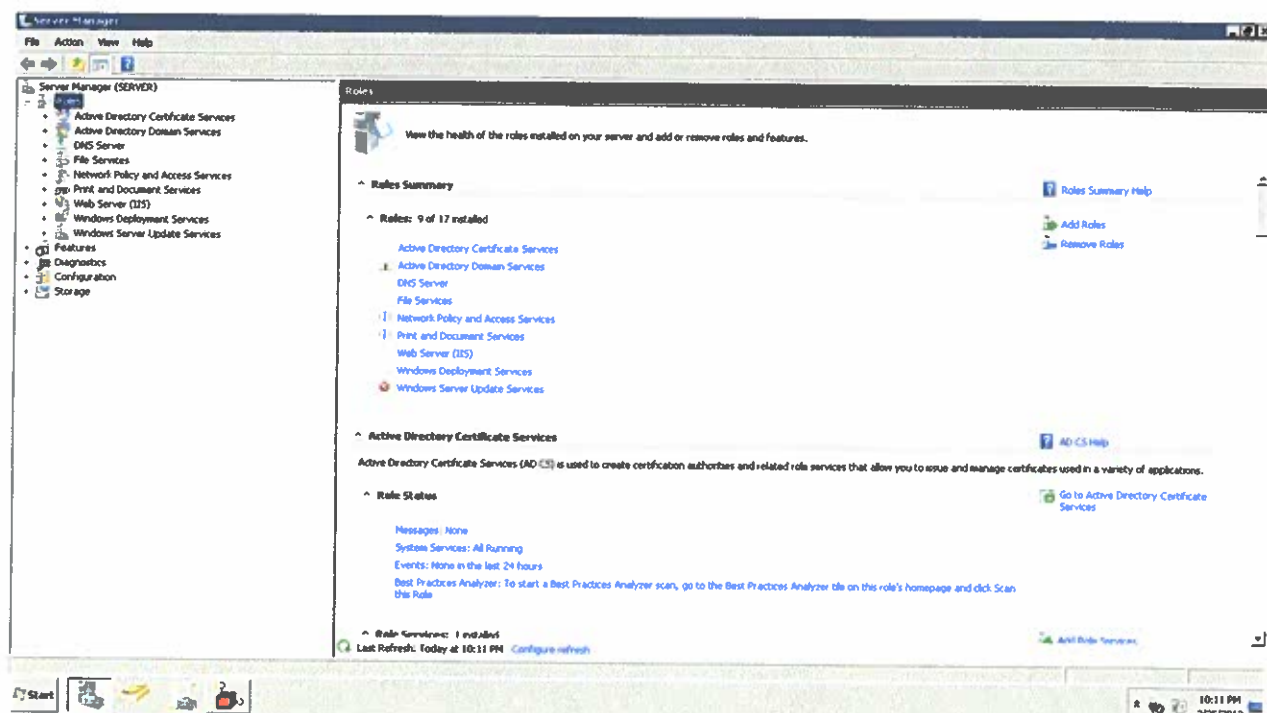
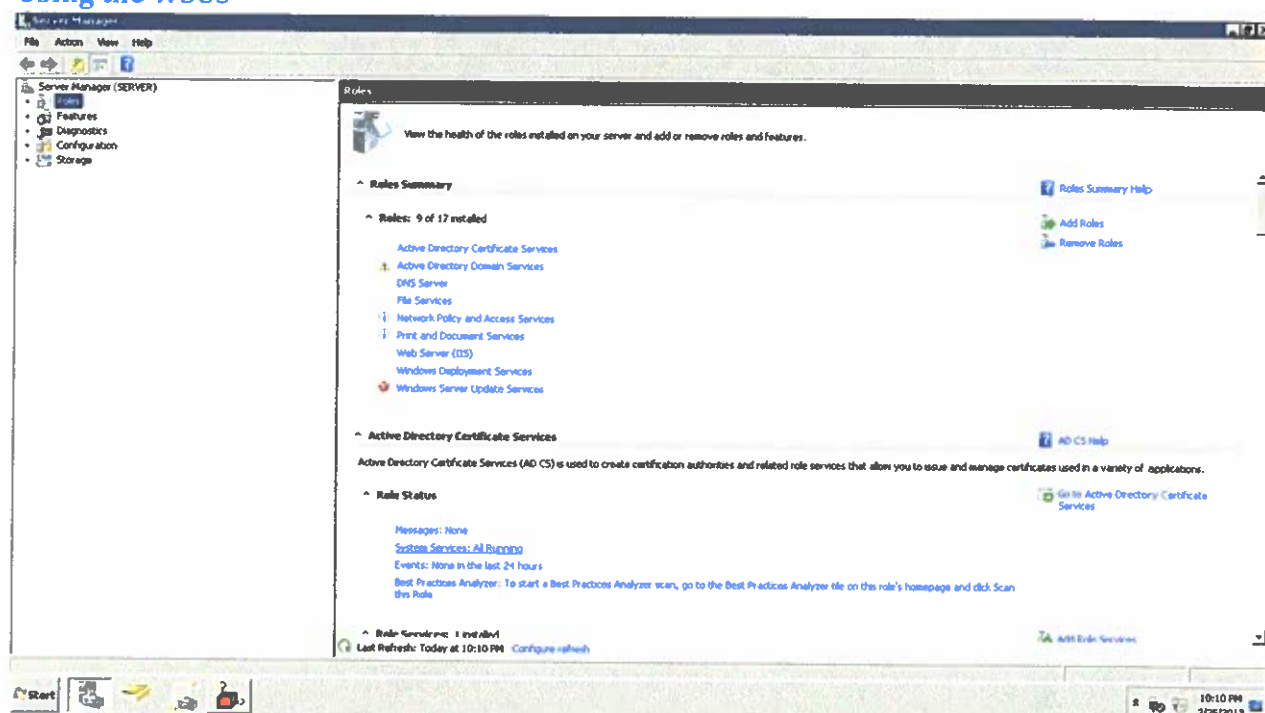
☐ Save Software Definition Package

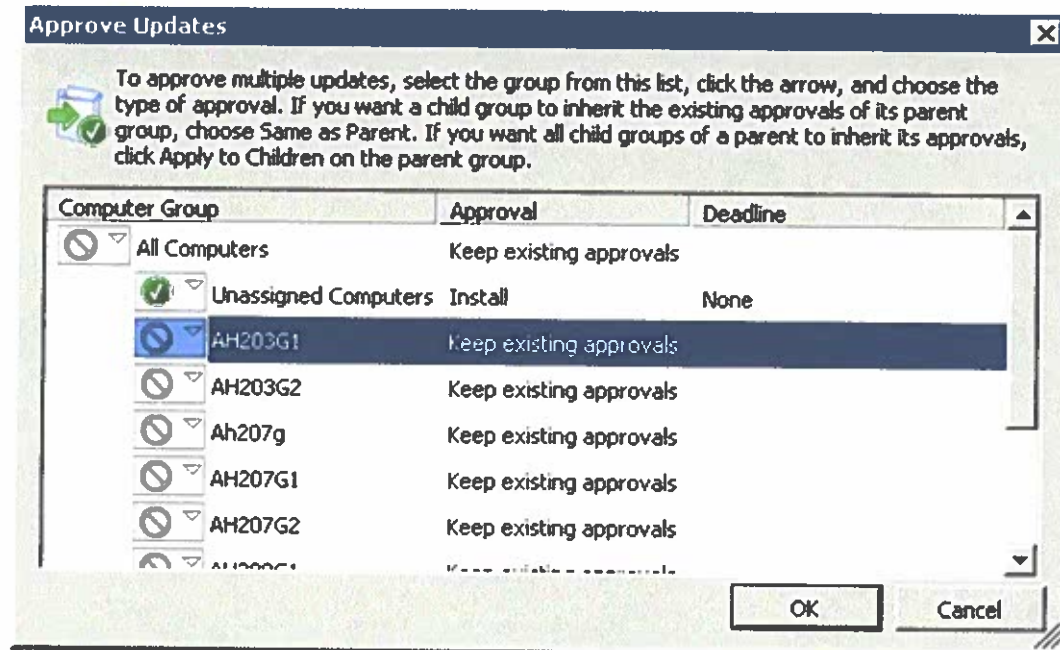
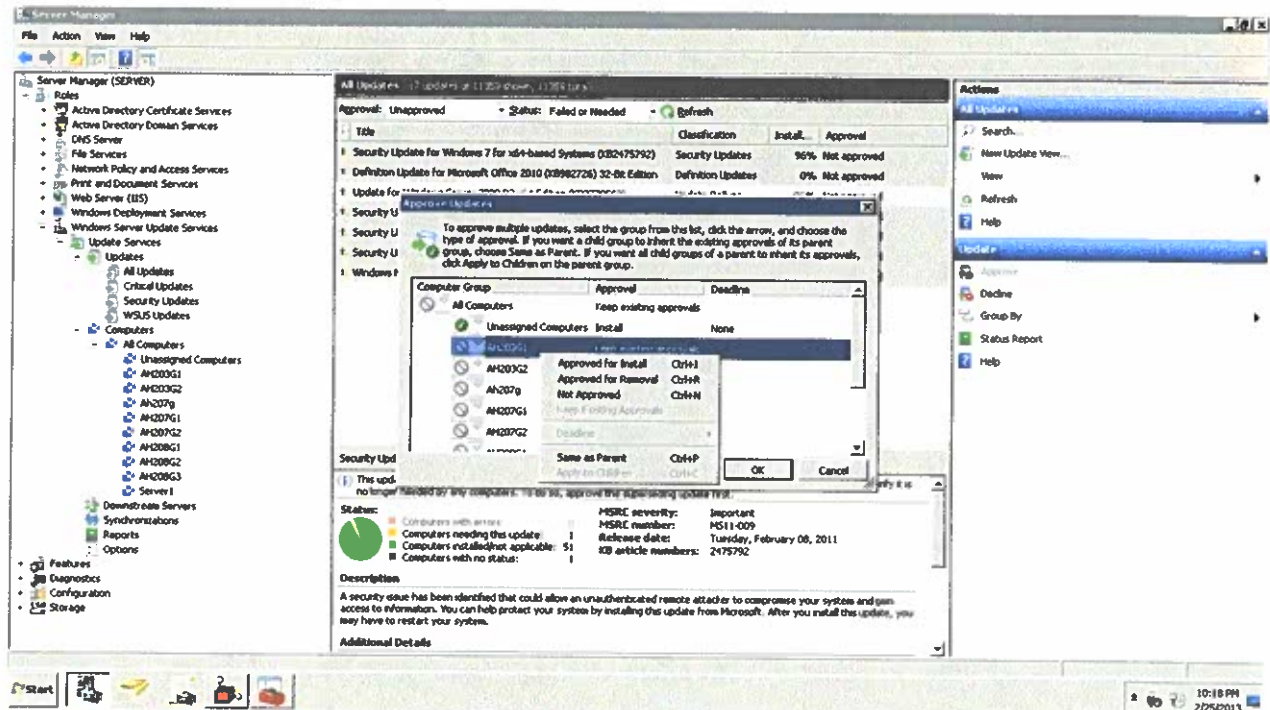
☐ Publish Metadata Only

Previous Next Cancel

Take all the defaults and approve it like a normal update under the updates tab in the Local Update Publisher.

Using the WSUS





Server Manager (SERVER)

File Action View Help

Roles

- Active Directory Certificate Services
- Active Directory Domain Services
- DNS Server
- File Services
- Network Policy and Access Services
- Print and Document Services
- Web Server (IIS)
- Windows Deployment Services
- Windows Server Update Services
- Update Services
 - Updates
 - All Updates
 - Critical Updates
 - Security Updates
 - WSUS Updates
 - Computers
 - Unassigned Computers
 - AH203G1
 - AH203G2
 - AH207g
 - AH207G1
 - AH207G2
 - AH208G1
 - AH208G2
 - AH208G3
 - Server1
 - Download Servers
 - Synchronizations
 - Reports
 - Options
- Features
- Diagnostics
- Configuration
- Storage

All Updates (17 updates at 11:20 AM, 11:20 AM)

Approvals: Unapproved Status: Failed or Needed Refresh

Title	Classification	Install...	Approval
Security Update for Windows 7 for x64-based Systems (KB2479552)	Security Updates	90%	Not approved
Definition Update for Microsoft Office 2010 (KB2479552) 32-bit Edition	Definition Updates	0%	Not approved
Update for Windows Server 2008 R2 x64 Edition (KB2479552)	Update Rollups	90%	Not approved
Security Update for Windows 7 for x64-based Systems (KB2479552)	Security Updates	90%	Not approved
Security Update for Microsoft .NET Framework 4 on Windows 7 (KB2479552)	Security Updates	90%	Not approved
Security Update for Windows 7 for x64-based Systems (KB2479552)	Security Updates	90%	Not approved
Windows Malicious Software Removal Tool (KB2479552)	Update Rollups	0%	Not approved

Security Update for Windows 7 for x64-based Systems (KB2479552)

This update is superseded by another update. Before you decline any superseded update, we recommend that you verify it is no longer needed by any computers. To do so, approve the superseding update first.

Status:

- Computers with errors: 1
- Computers needing this update: 1
- Computers installed/not applicable: 51
- Computers with no status: 1

MSRC severity: Important

MSRC numbers: MS11-009

Release dates: Tuesday, February 08, 2011

KB article numbers: 2479552

Description

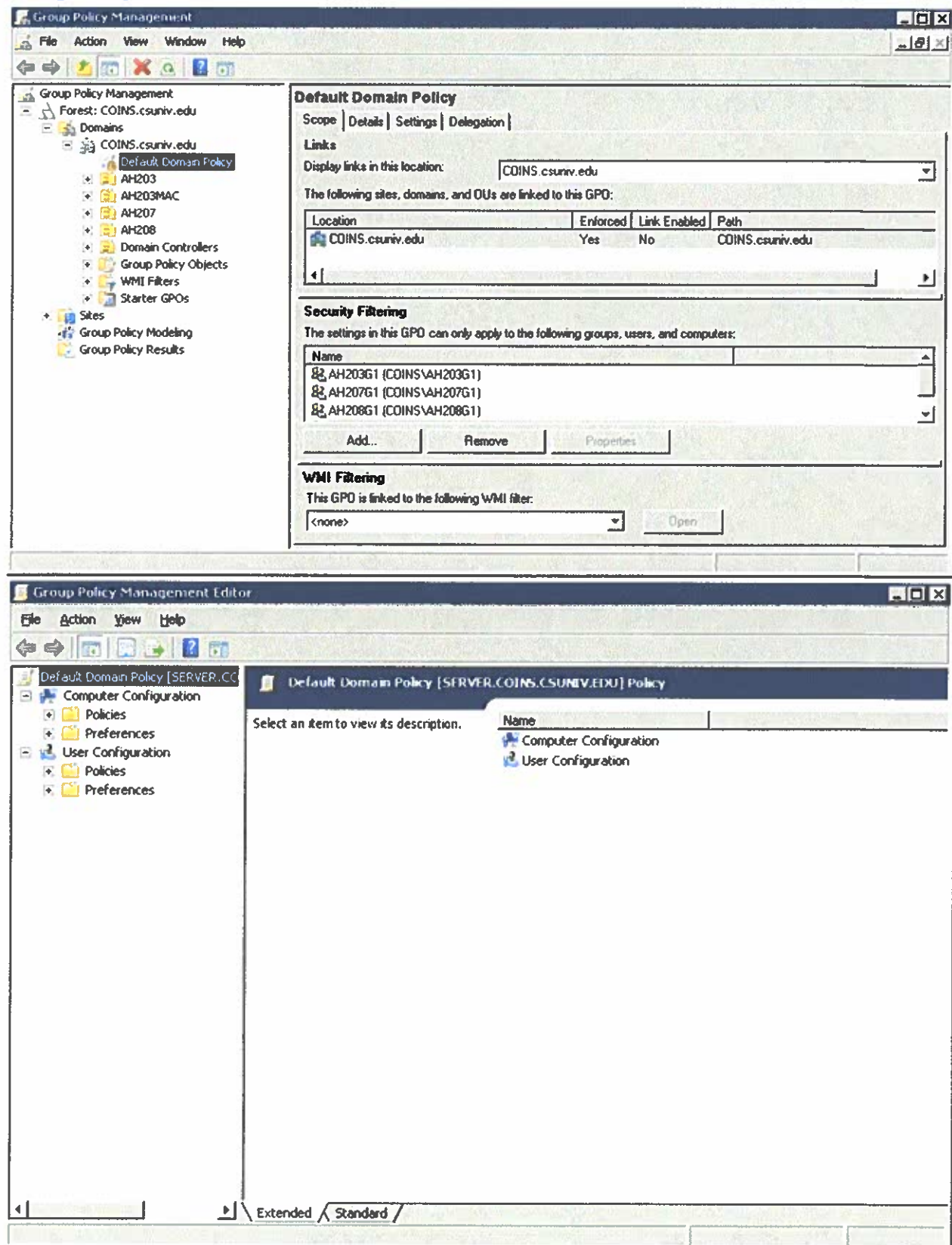
A security issue has been identified that could allow an unauthenticated remote attacker to compromise your system and gain access to information. You can help protect your system by installing this update from Microsoft. After you install this update, you may have to restart your system.

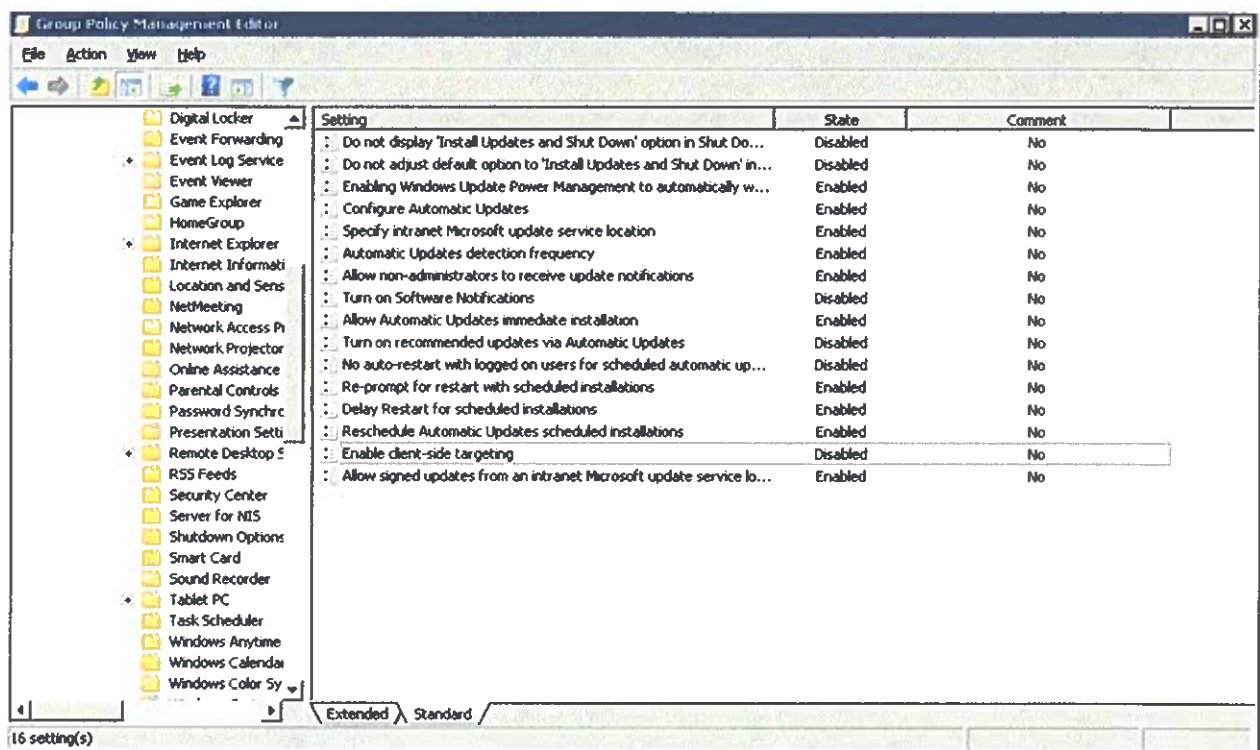
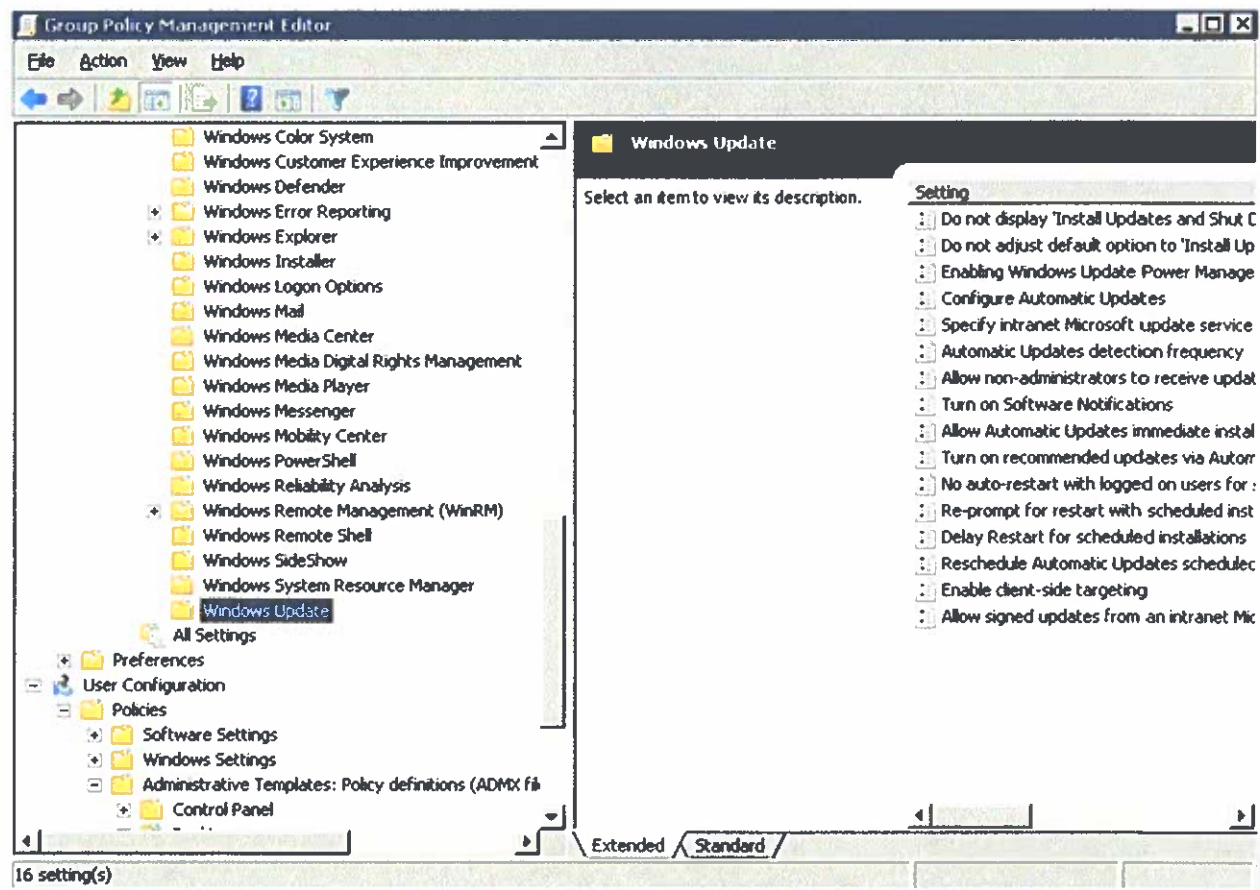
Additional Details

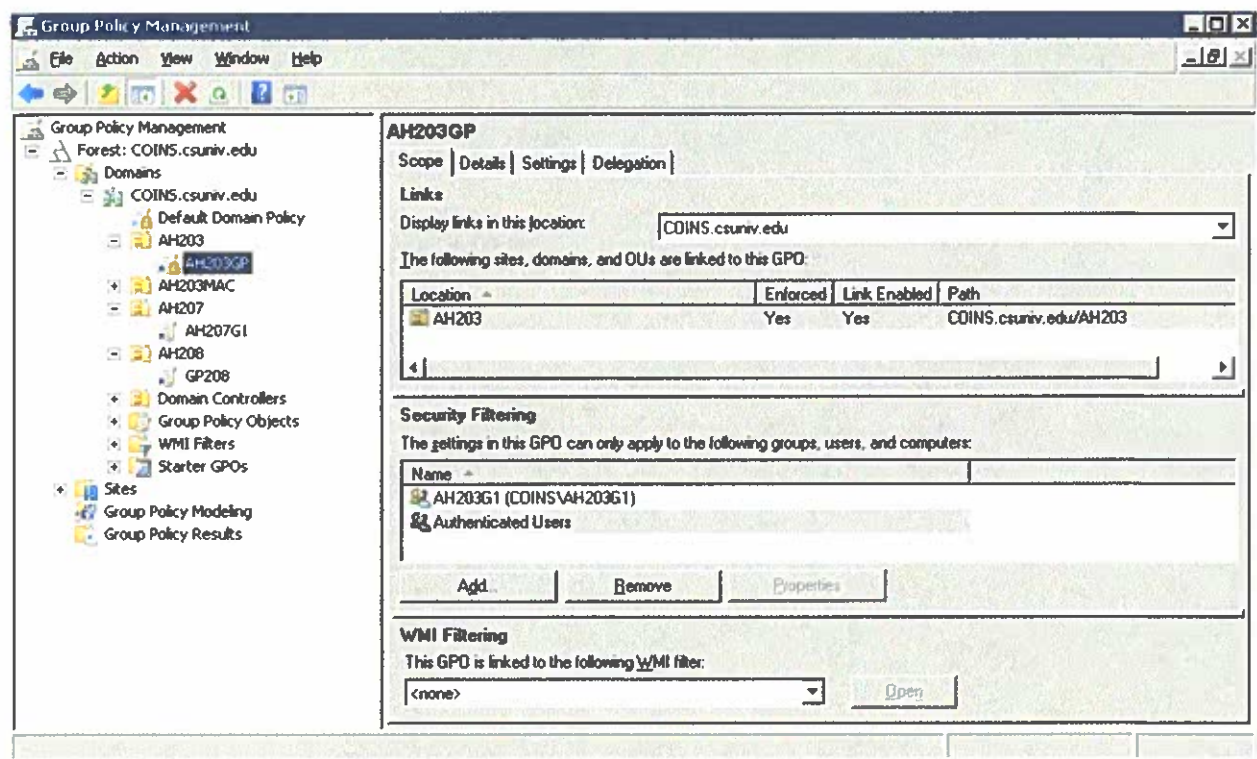
Actions

- All Updates
- Search...
- New Update View...
- View
- Refresh
- Help
- Update
- Approve...
- Decline
- Group By
- Status Report
- Help

Using Group Policies







This can be done for the policies in the OU and the Default Domain Policy

Using Print Server

Server Manager (SERVER)

Roles

- Active Directory Certificate Services
- Active Directory Domain Services
- DNS Server
- File Services
- Network Policy and Access Services
- Print and Document Services
 - Print Management
 - Custom Filters
 - All Printers (10)
 - All Drivers (15)
 - Printers Not Ready
 - Printers With Jobs
 - Print Servers
 - server (local)
 - Drivers
 - Faxes
 - Ports
 - Printers
- Web Server (IIS)
 - Windows Deployment
 - Windows Server Update Services
- Features
 - Diagnostics
 - Configuration
 - Storage

Printers

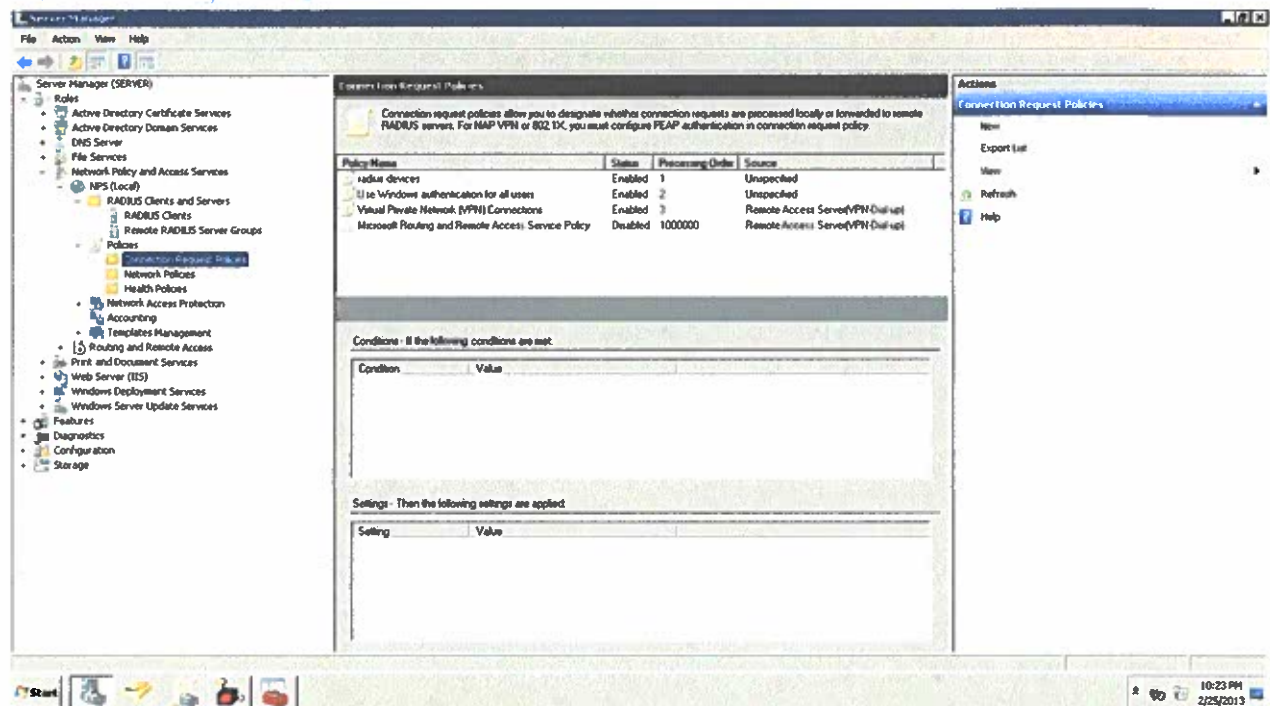
Printer Name	Queue Status	Jobs In...	Server Name	Driver Name
HP Brother H4-5280DW series	Ready	0	server (local)	Brother H4-5280DW series
HP Canon MF4360-4390	Ready	0	server (local)	Canon MF4360-4390
HP Fax (redirection 2)	Ready	0	server (local)	Microsoft Shared Fax Driver
HP LaserJet P4014/P4015 PCL6 o...	Ready	0	server (local)	HP LaserJet P4014/P4015 PCL6
HP LaserJet P4015	Ready	0	server (local)	HP LaserJet P4014/P4015 PCL6
HP Officejet 6500 E709a Series (...)	Ready	0	server (local)	HP Officejet 6500 E709a Series
HP Officejet 183 0-HP Officejet 6500 E70...	Ready	0	server (local)	HP Officejet 6500 E709a Series
HP Microsoft XPS Document Writer	Ready	0	server (local)	Microsoft XPS Document Writer
HP Microsoft XPS Document Writer (...)	Ready	0	server (local)	Microsoft XPS Document Writer
HP Send to OneNote 2010	Ready	0	server (local)	Send to Microsoft OneNote 2010 Dm...

Jobs

10:22 PM 2/25/2013

You can right click on any of the printers to see what is in the queue to print.

Network Policy Server



You can create and edit Policies or add radius clients by right clicking and following the wizards

Works Cited

"Network Policy Server." *Microsoft Tech Net*. Microsoft, n.d. Web.

"Preparing Network and Server for WSUS 3.0." *Microsoft Tech Net*. N.p., n.d. Web.